Inside the simulation

A virtual environment with a real warning engine. A glasses pathway with explicit engineering limits.

MODEL THE WORLD.
DO NOT MODEL AWAY THE LIMITS.

This brief explains how the browser turns geometry into one range reading, how the phone engine turns readings into cues, and what smart glasses would change. It assumes basic software and distance/time arithmetic, not specialist wearable knowledge.

Use this as a map

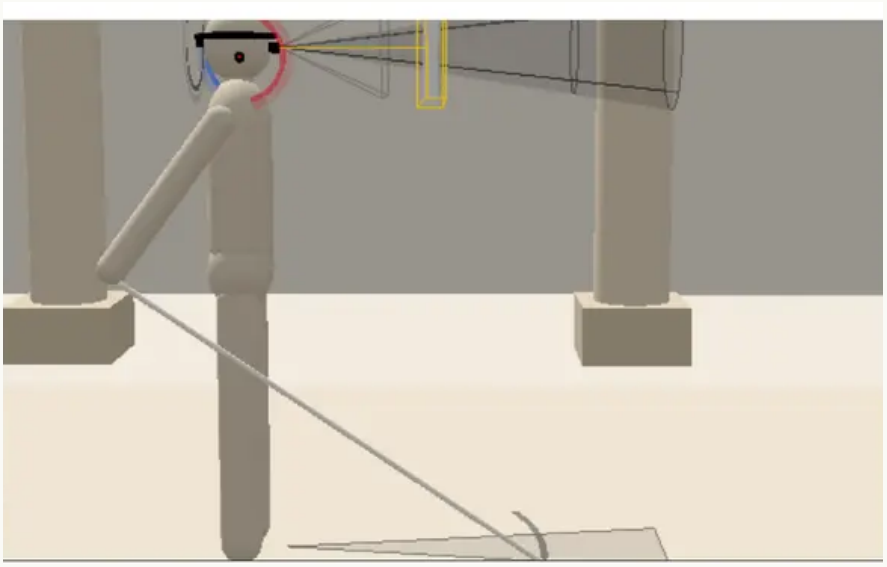
02 Read the running demo
03 Current wearable hardware
04 Software + clock boundaries
05 Sensor model and blind spots
06 Recorded engine evidence
07 TTC and latency
08 Failure-state semantics
09 Glasses integration architecture
10 Crop/resample versus real optics
11 Privacy + cloud implications
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How to read the evidence

CODE = implemented behaviour; **OBSERVED** = recorded local run; **MODEL** = calculation or sensor assumption; **PROPOSED** = not integrated. Blue denotes transport, teal local processing, amber optional ML/research. Dashed outlines mark proposed routes. Sources [1–10] are on page 13.

Read the demo

The scene illustrates a head-height obstacle. The gauge and score expose what the engine actually did—not what the animation author wanted it to do.



OBSERVED UI

READY

NEAR ahead

1 Course + expected limit

2 Real-engine state and alert

3 Raw range versus smoothed urgency

4 Separate sound and vibration staves

Fig. 1. The cane can pass below the sign while the brow-mounted sensor faces it. Scene crop from the actual production-demo screenshot; the enlarged status at right transcribes the observed state. This is software rendering, not a physical walk. [1, 3]

Seven courses make different claims

Course	Question it makes visible
Sign	Can the range policy warn about a head-height object the cane passes under?
Scaffold / pedestrian	What is missed beside the cone? What changes with relative closing speed?
Dark branch	What happens when the assumed material return range is short?
Glass	Does an explicitly unknown measurement remove the stale alert and degrade state?
Plaza / stairs	Can we explain silence without claiming that the path is clear?

What is actually worn?

Current headband architecture—not glasses hardware. The drawing is a functional placement sketch, not a measured assembly or wiring CAD. [1, 2]

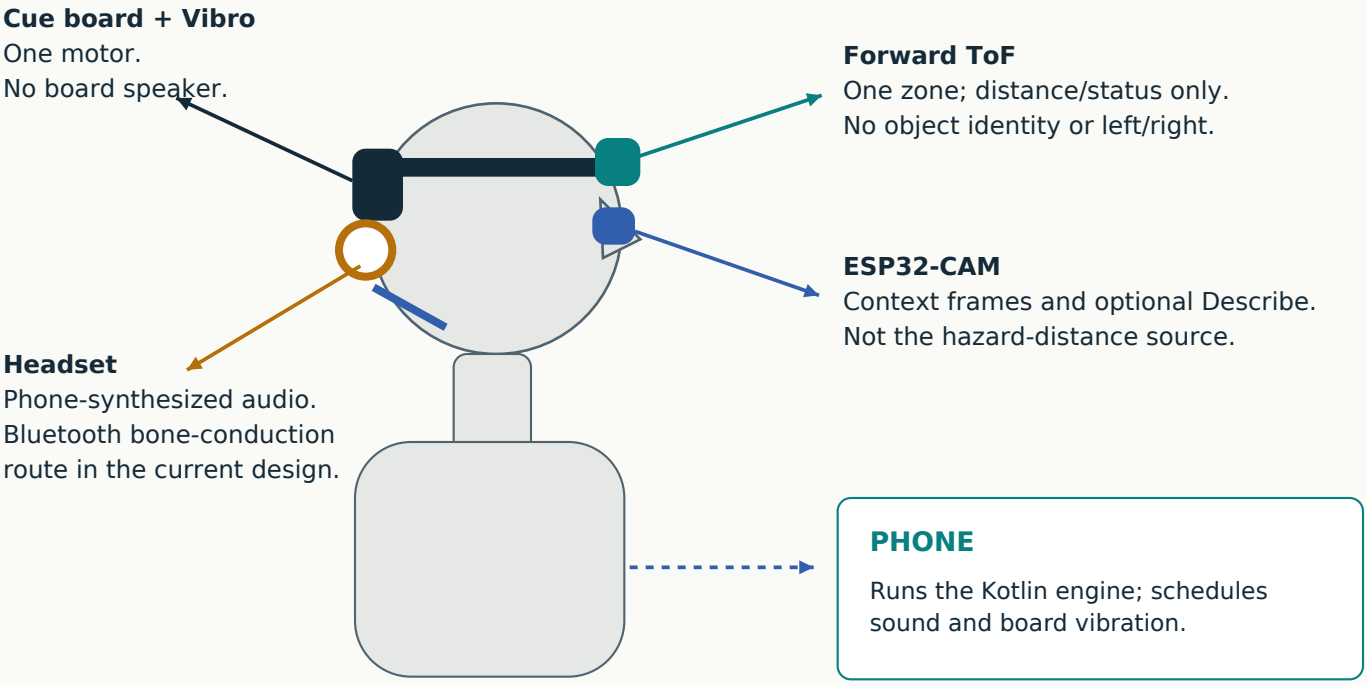


Fig. 2. Sensing, deciding and actuating live on different devices. Losing the cue board cannot be repaired by telling that unpowered board to vibrate; phone-side failure audio and the powered board watchdog cover different failures. Sketch not to scale. [1, 2, 4]

Why keep the cane?

The sensor looks near head height through a narrow cone. A drop-off or low obstacle may never intersect it. The cane samples a different part of the environment, so this design supplements it rather than establishing obstacle-free travel.

The one-motor consequence

Wire fields named `hl` and `hr` do not create two actuators: the v2 board applies their maximum to one Vibro motor. The range is forward-only, so a cue must not invent a safe steering direction.

How a warning crosses the software

The browser supplies a scene and simulated measurements. The existing phone engine remains the authority for warnings. [1, 2, 4]

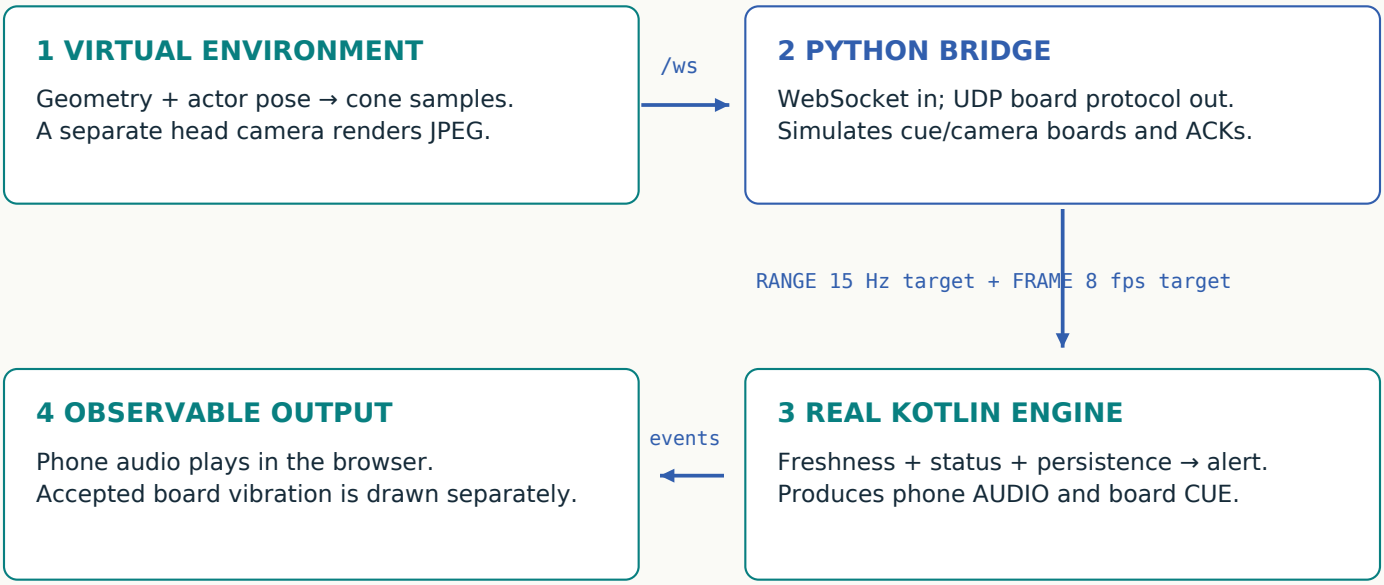


Fig. 3. The simulator exercises the same warning policy through real protocol boundaries. Its loopback ACK is evidence of software/protocol behaviour, not transducer onset or Bluetooth audio delay. Nominal rates are not guaranteed under render load. [1, 4]

Three clocks must not be silently subtracted

Clock	What it means / how to use it
Course tS	Choreography position; may pause or reset. Sensor rows carry the latest course event time, not a synchronized capture instant.
wallMs	Recorder receipt time (epoch ms). Same-receiver intervals are useful, but they include transport/scheduling.
Board / engine time	Source timestamps used for freshness and policy timing. Compare across devices only after clock-offset handling.

Where to inspect: `scene/world.ts` → `link/link.ts` → `tools/sim/world_bridge.py` → `HazardPolicy.kt`. The bus carries observations and commands; it is not a second hazard policy.

How the virtual sensor sees

Time-of-flight (ToF) estimates distance from reflected light. Here it is approximated with geometric rays and assumed material response—not a physical optical simulator.

TOP VIEW · ONE FORWARD ZONE

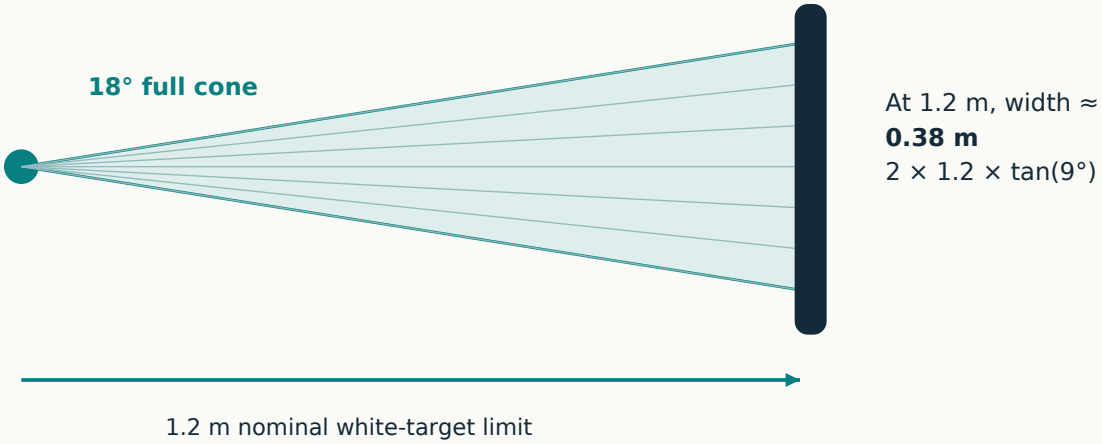
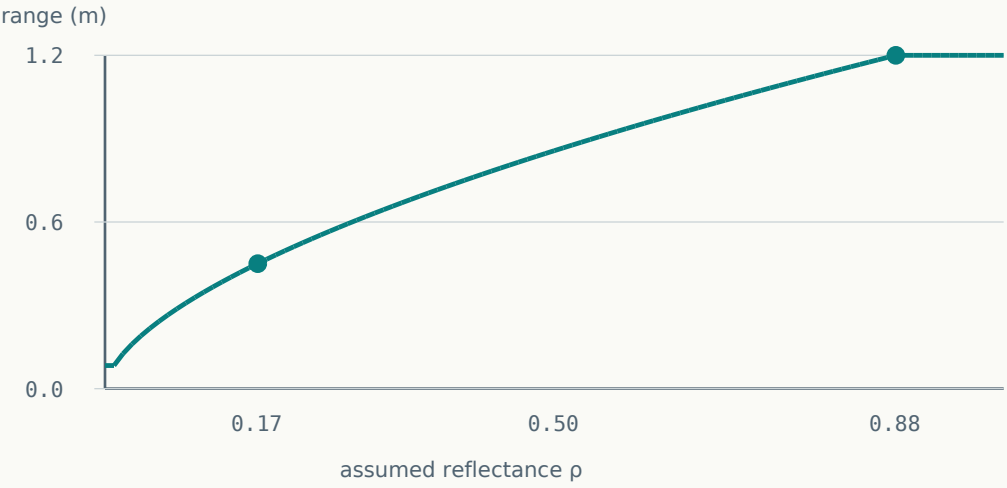


Fig. 4. A circular 18° cone is narrow even at its outer range. The browser samples a centre ray plus three rings of twelve rays (37 total); these are numerical samples of one zone, not 37 hardware pixels. Model geometry, not a measured field of view. [1, 2]

For each ray the code intersects yaw-rotated boxes, retains the nearest forward projection, and requires at least four rays to hit geometry. This coverage rule is an assumption. No target produces status 1 with mm = 0; it does not mean a surface is at zero distance.



MODEL
dmax = 1200 × $(\rho / 0.88)^{0.596}$ mm

Capped at 1200 mm.
Noise σ = 4 + 0.005d mm.

Fig. 5. The fit reproduces two VL53L4CD datasheet anchors: 0.45 m grey and 1.20 m white. Applying that fit to VL53L4ED, assigning reflectance by object kind, and using a four-ray threshold are uncalibrated assumptions. σ is 10 mm at d = 1200 mm. Bench data must replace them. [2, 8]

A real engine trace, not a scripted warning

The sign episode below came from the production browser → bridge → installed Kotlin CLI. Approach escalation was OFF. Dots and band changes are loaded from the saved event log. [3]

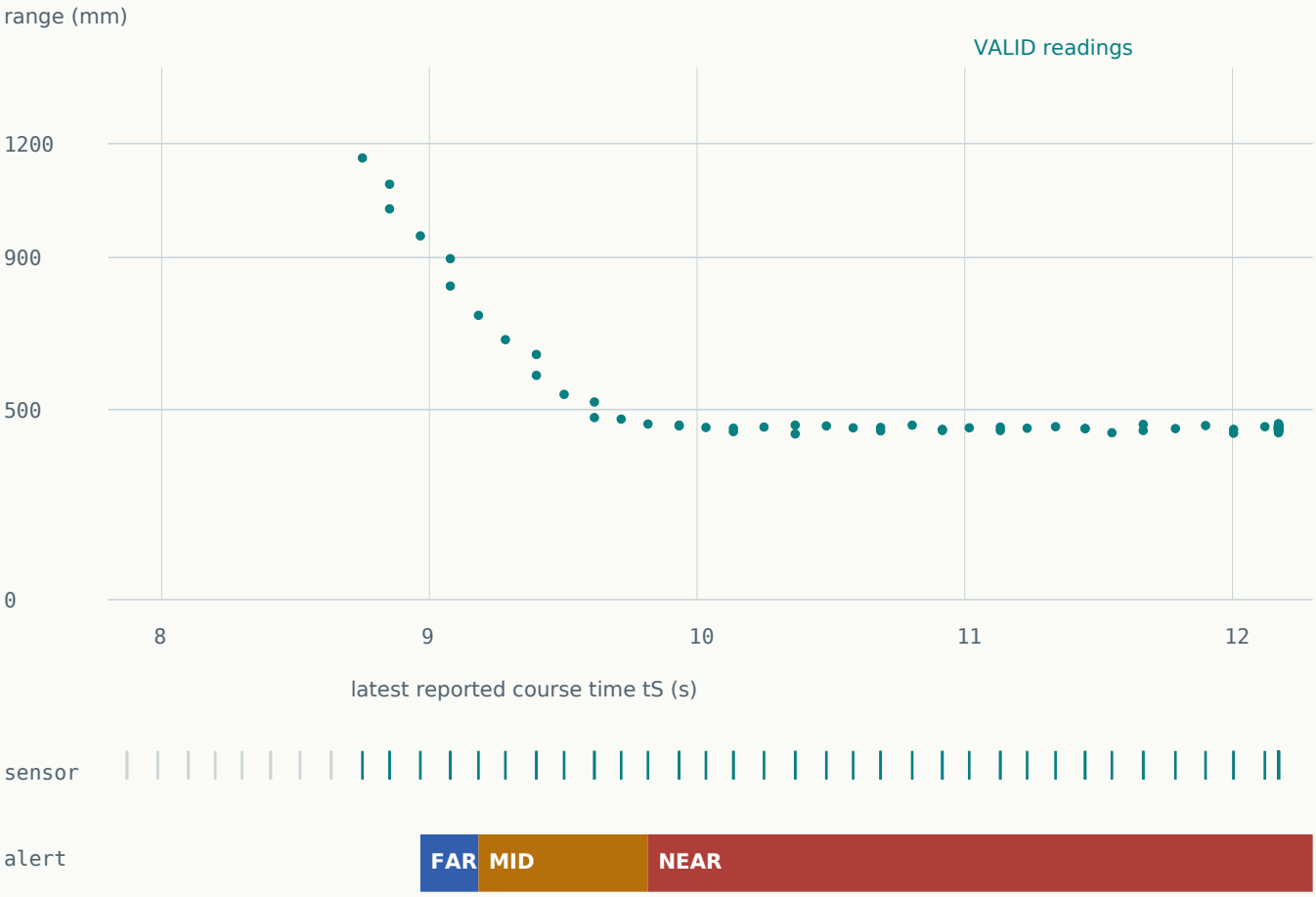


Fig. 6. 206 sensor rows and 83 non-null alert rows were recorded; the plot shows the 7.8–12.3 s approach/stop window. NO_TARGET is the grey categorical strip, never a 0 mm obstacle. Band changes are real engine events, not threshold lines drawn as fake alerts. [3]

> **235 ms was observed between the recorder receiving the first VALID sample and receiving the first FAR alert.** This is a local application receipt interval, including policy persistence and scheduling. It is not sensor-to-ear latency, an acoustic measurement, or a certified response time.

Check the interpretation: the walker stops near 0.45 m and NEAR continues. A stopped animation is not proof that an independent user avoided a collision; the trajectory was scripted. The saved trace preserves its tuning and source provenance.

Urgency and the latency budget

A faster approach reduces the time available even when the current distance is identical. The existing experimental rule estimates closing speed from VALID samples. [4]

Distance is a position; time-to-contact includes motion

With approximately constant positive closing speed v, travel distance satisfies d = v × t. Rearranging gives **TTC = d / v**. Metres divided by metres/second gives seconds; at zero closing speed there is no finite forward-contact estimate. The engine uses a least-squares slope over a 300 ms window, needs at least three VALID samples, and clears the estimate on a missing target.

SAME 800 mm · FAST APPROACH

v = 800 mm/s → TTC = 1.0 s.
v ≥ 300 mm/s and TTC < 1.5 s.
Experimental rule raises MID → NEAR.

SAME 800 mm · SLOW APPROACH

v = 200 mm/s → TTC = 4.0 s.
Speed and TTC criteria are not met.
No approach escalation; MID remains.

Fig. 7. The examples change speed, not distance. They are calculations, not recorded test cases. Approach escalation can advance urgency after a return exists; it cannot see an unseen obstacle. At 1.0 m/s the 1.2 m white-target envelope offers at most 1.2 s before contact, before every delay below. [4]

Where that geometric time is spent

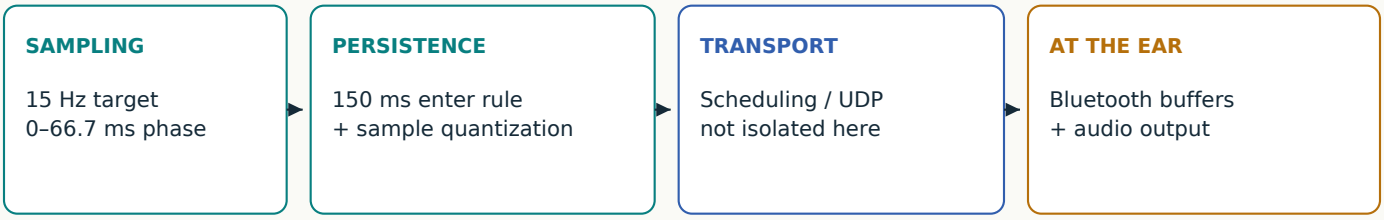


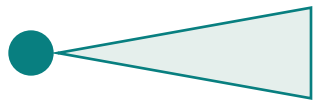
Fig. 8. A protocol ACK is not the final block. Bluetooth SIG literature gives typical A2DP delays around 100–200 ms, but that is not a measurement of this headset or of Ray-Ban Meta. A total needs a shared-clock event marker and a microphone/contact measurement at the actual output. [4, 7]

Model implication: 15 Hz and 150 ms do not imply exactly 216.7 ms to warning. Sample phase, entry evidence, onset gating, processing and output routing all matter; TTC also changes policy behaviour. Report distributions and failures, not one optimistic constant.

Failure is a state, not silence

“No warning” has several causes. The interface must keep them distinguishable because each implies a different next action. [1, 4]

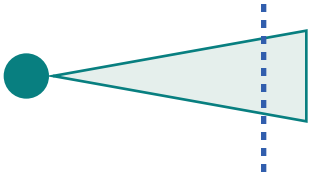
OPEN PLAZA



NO_TARGET

No usable surface in range.
Do not infer a clear path.

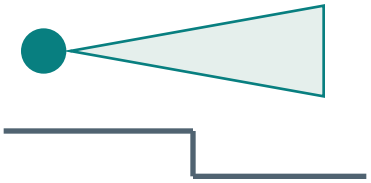
GLASS FIXTURE



UNKNOWN

Model deliberately invalidates the measurement. Not a glass detector.

STAIRS DOWN



NO_TARGET

Drop-off is below the cone.
The cane still matters.

Fig. 9. Identical quietness can mean a scene beyond range, a known sensing failure, or a hazard outside the zone. Only the explicit glass fixture supplies UNKNOWN here; physical glass can also return misleading ranges or no target. Schematics not to scale. [1, 2]

Availability is stricter than the distance policy

READY
Required liveness and usable range state hold.

→

DEGRADED
ToF stale or UNKNOWN; no invented replacement range.

→

UNAVAILABLE
Frame/heartbeat stale, ACK failure, route loss...

Illustrative failure progression; recovery is possible when guards hold.

In the current monitor, frame age, board heartbeat age and range age each have a 500 ms default limit. Range trouble degrades state; a stale camera frame makes the system UNAVAILABLE rather than silently entering a ToF-only mode. Camera content does not set the hazard band, but camera freshness still gates availability.

> **Failure proof shown in the local run:** unplugging the cue board produced range_stale, then cue_beat_missing; reconnecting recovered READY. A powered cue board's local watchdog and the phone's failure audio are independent paths. An unplugged board cannot provide its own alert.

What changes with smart glasses?

The useful glasses proposal is a different camera source and possible output route—not “replace the entire headband”. This pathway is not integrated in the current code. [5, 6]

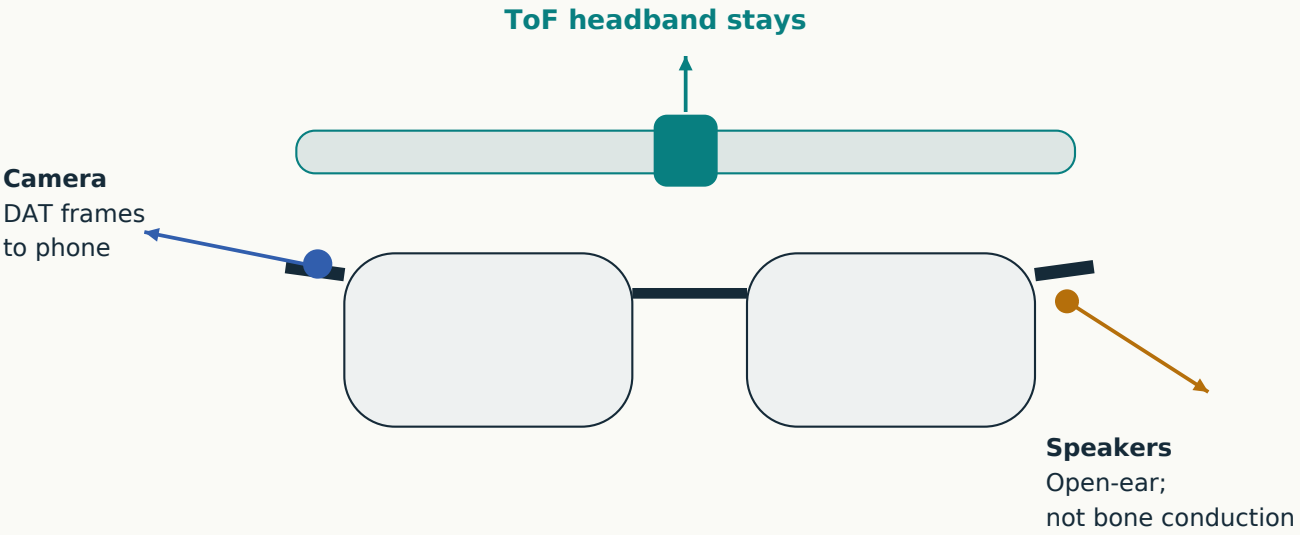


Fig. 10. The near-field ranging module and haptic board remain separate from the glasses. Camera frames and media audio use different APIs/routes. Generic functional sketch, not Ray-Ban CAD or a claim about exact component positions. [5, 6]

GLASSES → PHONE

DAT camera registration, permission, session lifecycle and timestamp handling.

PHONE → GLASSES

Ordinary Bluetooth A2DP output. Verify the actual route and audible onset.

Do not confuse the profiles. A2DP is high-quality output. HFP enables microphone input but changes output to mono voice quality. DAT also documents in-stream microphone audio in development/beta channels; that is not production support for an unrestricted audio feature. [6]

Product implication. Open-ear glasses speakers do not satisfy a bone-conduction requirement by equivalence. Keep the existing headset path unless the alternative route is explicitly accepted and evaluated for intelligibility, environmental audibility and failure announcements.

Geometry implication. A different mount changes camera-to-ToF alignment. Never fuse a camera object and the one range reading by identity without a calibrated, separately validated association method.

Pixels are not optical evidence

The built Meta profile studies an image-interface shape. It never changes the live bridge, ToF, engine, camera pacing or audio route. [1]

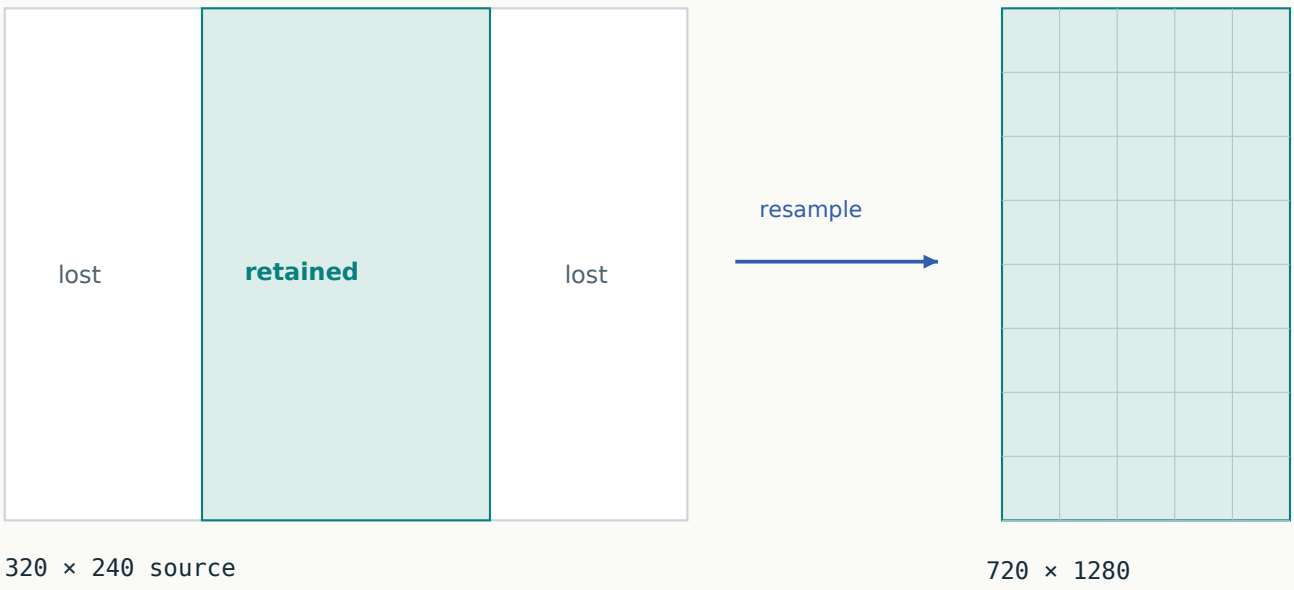


Fig. 11. The crop is 135 × 240 source pixels at x = 92.5, y = 0; resizing expands it by 5.33× on each axis. It retains 42.2% of the source width and adds no optical detail. Diagram shows image geometry, not a captured glasses view. [1]

The crop follows aspect ratio: width = 240 × 9/16 = 135 pixels. Centre offset = (320 − 135)/2 = 92.5 pixels. Selecting 720 × 1280 therefore proves an output dimension—not a field of view, effective resolution, calibrated mount or SDK connection.

What a real DAT path still needs

Permissions and lifecycle: app registration, Bluetooth/device access, camera permission, a started session, stream state handling and revoked-permission/disconnect behaviour. Mock Device Kit can test application lifecycle logic, not real RF reliability or ear latency. [5]

Power and heat: sustained capture, encoding and radio traffic can trade battery life and phone temperature for image quality. Measure actual current and runtime; this simulation has no validated energy or thermal model.

Bandwidth and freshness: documented rates are 2, 7, 15, 24 or 30 fps; HIGH is 720 × 1280. DAT adapts compression/quality to Bluetooth bandwidth. Decoded RGB at 30 fps is about **82.9 MB/s** inside the app—not a Bluetooth bitrate. At 2 fps the nominal 500 ms inter-frame interval already consumes the current freshness limit; jitter makes it unsuitable without an explicitly reviewed liveness design. [4, 5]

Describe, privacy and cloud placement

Describe is optional context, not a sensor or an authority that the path is safe. A slow or failed description must not become an invented warning or all-clear. [1, 9]

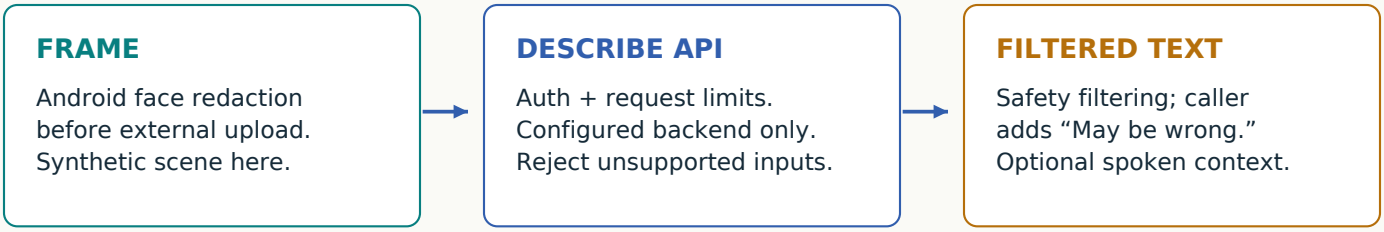


Fig. 12. A cloud description crosses a privacy boundary before the model runs. A server-side blur pass would not erase a pre-upload exposure; phone-side redaction is a distinct control. No real cloud model was invoked in the documented local smoke. [9]

Why the AWS/Annapurna connection is defensible

Warning deadlines constrain placement: a network round trip has no reliable bound tight enough to make it a dependency for urgent cues. Optional scene descriptions can wait and can therefore be evaluated on a real configured inference host. The simulator supplies controlled scenes and captures; the batch CLI compares explicit human-labelled objects with returned text, timings, failures and declared backend metadata.

A Neuron tutorial genuinely documents Llama 3.2 Vision on `inf2.48xlarge` or `trn1.32xlarge`. That is not the project's small `inf2.xlarge` SmolVLM path. The tutorial is versioned 2.27.1; verify current SDK support, quota, model access and compile before promising a deployment. [10]

What a trustworthy receipt can—and cannot—say

Record observed JPEG bytes, redaction outcome and verified provider/model fields; label missing fields unknown. “The API contract stores no images” is a policy statement, not a measured zero-second retention guarantee across every host, log and provider. Synthetic faceless avatars also do not validate face detection on real bystanders.

> The implemented Describe evaluator is a limited English lexicon audit, not a semantic safety proof. Failures remain visible and zero denominators stay null. Cone-only truth cannot label all camera-visible objects; full-image labels require independent annotation.

Turn the demo into research

A convincing demo makes an observable claim and also shows the experiment that could falsify it. Promotion to user-facing mobility claims requires evidence beyond this virtual world.

1. **1 SOFTWARE LOOP**
   Repeat scene + policy + failure traces; export source clocks and tuning.
2. **2 SENSOR BENCH**
   White/grey/dark/glass targets; vary angle, distance and ambient light.
3. **3 OUTPUT TIMING**
   Measure acoustic headset onset and motor onset, not only ACK receipt.
4. **4 EXTERNAL REPLAY**
   Aria calibration + depth projection; separate ideal-depth from ToF assumptions.
5. **5 SUPERVISED STUDY**
   Orientation-and-mobility review, safeguards, and predeclared outcomes.

Fig. 13. Each rung tests something the previous rung cannot: software logic, physical returns, physical actuation, external data generalization, then supervised human benefit. ADT sequence/derived public displays need licence review and permission; no Aria assets were included in this demo. [1, 3, 8]

A 90-second stage sequence

0–25 s: show the sign, range, real engine and separate audio/vibration score. **25–45 s:** switch to glass or stairs and explain the blind spot. **45–60 s:** unplug/recover a board so failure is visibly announced. **60–90 s:** show an exported episode and the glasses crop sidecar; say which integration is still proposed.

> **Next experiment:** freeze a paired baseline-versus-approach protocol before running it. Hold paths, speeds and sensor streams constant; define washout periods, miss/false-alarm labels and promotion criteria. The camera-frame crop is not independent ground truth. Confirm demo and data rules with HackTX organizers before the submission.

Sources, glossary and reproduction

Sources support specific claims, not blanket endorsements. Repository paths refer to this delivered workspace; upstream engine code is pinned to __UP7__.

1 **Implemented world + profile boundaries** — world/README.md; world/README-profiles.md; docs/research/WORKLOG.md

2 **Sensor model and material assumptions** — world/src/sensor/tof.ts; world/src/scene/world.ts

3 **Recorded evidence included with this PDF source** — assets/sign-trace.json; 872 original JSONL rows, 206 sensor rows; screenshot assets/simulation.png

4 **Upstream policy, tuning and availability** — HazardPolicy.kt; Tuning.kt; SystemMonitor.kt (phone/core)

5 **Meta DAT Android integration guide** — Registration, permissions, stream lifecycle, dimensions/rates and adaptive compression. Retrieved 24 Oct 2026.

6 **Meta: microphones and speakers** — A2DP/HFP routing; beta in-stream capture; route verification. Retrieved 24 Oct 2026.

7 **Bluetooth SIG: Introducing Bluetooth LE Audio** — Nick Hunn (2022), pp. 128–129: typical A2DP delay; not a measurement of Determine hardware.

8 **Sensor anchors and Aria use restrictions** — ST VL53L4CD datasheet DS13812; ADT sequence licence §1–3. L4CD → L4ED extrapolation is MODEL, not validation. ST datasheet.

9 **Describe contract and evaluation tool** — cloud/API.md; tools/sim/EVALUATION.md; docs/research/annapurna-labs.md

10 **AWS Neuron 2.27.1: Llama 3.2 multimodal** — Versioned tutorial; documented trn1.32xlarge / inf2.48xlarge setup.

Working vocabulary

ToF: reflected-light ranging. **TTC**: time-to-contact under a closing-speed model. **DAT**: Meta Wearables Device Access Toolkit. **ACK**: protocol acknowledgment, not physical output. **VLM**: vision-language model. **SIL**: software-in-the-loop, not hardware validation.

Reproduce / edit

Build the demo using `world/README.md`. Rebuild this PDF with `python3 docs/visual-guide/build.py` after installing its requirements. Source, snapshot data, screenshot and calculation metadata live beside this PDF. Body text and diagrams remain editable; PDF outlines and cover contents are clickable.